CS 3120 Exam 1

This packet contains exam 1. This **cover sheet** is here to provide instructions, and to cover the questions until the quiz begins. **do not open this quiz packet** until your proctor instructs you to do so.

You will have 1 hour and 15 minutes to complete this exam. Each quiz is two pages (front and back of one sheet of paper) worth of questions. Make sure to **write your name and computing id at the top of each individual page**.

When you are finished, simply submit this packet at the front of the classroom.

This quiz is CLOSED text book, closed-notes, closed-calculator, closed-cell phone, closed-computer, closed-neighbor, etc. Questions are worth different amounts, so be sure to look over all the questions and plan your time accordingly. Please sign the honor pledge below.

You step in the stream,
But the water has moved on.
This page is not here.

Module 0: Computers and Cardinality

Name ______________________________

1. [6 points] Answer the following True/False questions.

For a computing model, it is allowed that $\Sigma = \mathbb{N}$	**True**	**False**
Computers (as we defined them) can only understand *binary strings*	**True**	**False**
$\lvert\{0,1\}^n\rvert > \lvert\mathbb{N}\rvert$ for some $n \in \mathbb{N}$	**True**	**False**
It is possible to implement every function (from strings to strings) in *python*	**True**	**False**
All infinite sets have an *injection* to $\mathbb{N}$	**True**	**False**
All infinite sets have a *surjection* to $\mathbb{N}$	**True**	**False**

2. [4 points] Consider $\mathbb{Q}$, the set of all *rational numbers*. *Recall that rational numbers can be expressed as fractions* $\frac{x \in Z}{y \in Z^+}$. Is there a *bijection* from $\mathbb{Q}$ to $\mathbb{N}$? Why or why not?

Module 2: Regular Languages

Name

3. [6 points] Answer the following True/False questions.

$\{1^*0^n1^n0^* \mid n \geq 0\}$ is a regular language because we can always set $n = 0$, reducing the language to just 1^*0^*	**True**	**False**
It is possible for a *DFA* to have at least one accept state ($\|F\| > 0$) and for the language of that *DFA* to be the empty set ($\emptyset$)	**True**	**False**
The *regular languages* are closed under *complement*	**True**	**False**
Any *regular language* can be expressed using an *NFA*, but some *regular languages* cannot be expressed with a *DFA* (an *NFA* is required)	**True**	**False**
All regular expressions can be represented using only the *union*, *concatenation*, and *star* operators	**True**	**False**
*DFA*s are required to have a *finite* number of states ($\|Q\| < \|\mathbb{N}\|$)	**True**	**False**

4. [2 points] Draw a working *DFA* (NOT an *NFA*) for the following language. *The language of binary strings that contain EXACTLY ONE occurrence of the substring 101*. For example, 000101001 should accept but 00101010 should not (note that the two instances overlap in this second case)

5. [2 points] Prove that the regular languages are closed under set difference. Recall that if A and B are languages of strings, then $A \setminus B = \{w \in \Sigma^* \mid w \in A \land w \notin B\}$.

Miscellaneous

6. [3 points] Write a regular expression for the language consisting of all strings over $\Sigma = \{a, b, c\}$ in which all occurrences of c appear *after* all occurrences of a. For example, *aaabbbccc* is in the language because all of the a's come before all of the c's, but *abca* is not in the language because there is at least one a after at least one c.

7. [3 points] Consider the following regular expression where $\Sigma = \{0, 1\}$: $(01 \cup (10)^*)$. In class, we showed a construction on how to convert any arbitrary regular expression into an equivalent *NFA*. Use this EXACT algorithm to convert this regular expression into an *NFA* and draw the result.

8. [3 points] Use the pumping lemma to prove the following language is NOT regular: $L = \{a^{2^n} \mid n \geq 0\}$. In other words, the language of strings that are a number of a's that are a power of 2 ($L = \{a, aa, aaaa, aaaaaaaa, ...\}$)